

Here is the complete command guide for **testing, model evaluation, training, and prediction** across both the **Elasticsearch Pipeline** (`main_es.py`) and the **CSV/MySQL Pipeline** (`main.py`).

1. Automated Test Suite (Pytest)

Run unit tests to verify data processing, metric calculations, and validation logic.

```
# Run all unit tests
python -m pytest tests/ -v

# Run specific test modules
python -m pytest tests/test_evaluation.py -v # Tests MAE, RMSE, MAPE & holdout splits
python -m pytest tests/test_forecaster.py -v # Tests model building & future dataframe generation
python -m pytest tests/test_ingestion.py -v  # Tests metric scaling, deduplication & validation
```

2. Elasticsearch Pipeline (`main_es.py`)

Queries metrics directly from `http://192.168.12.94:9200` .

A. Data Inspection & Dataset Verification

```
# Fetch 7 days of 1-min data, output summary stats, and export CSV + Excel
python main_es.py inspect

# Inspect 14 days of data with custom core normalization (e.g. 8 cores)
python main_es.py inspect --days 14 --cores 8
```

B. Model Evaluation (MAE / RMSE / MAPE Accuracy Testing)

Evaluates model accuracy by splitting historical data into training and a trailing holdout test set.

```
# Evaluate 1-minute model (2-hour holdout window)
python main_es.py evaluate

# Evaluate hourly model (24-hour holdout window)
python main_es.py evaluate --resolution 1h

# Evaluate daily model (2-day holdout window)
python main_es.py evaluate --resolution D
```

C. Model Fitting (Training)

```
# Train default 1-minute model (60-minute lookback)
python main_es.py fit

# Train hourly model
python main_es.py fit --resolution 1h

# Train with 14 days of historical lookback
python main_es.py fit --days 14
```

D. Prediction & Short-Term Forecasting

```
# Predict 60 minutes ahead and output chart + CSV
python main_es.py predict

# Predict 120 minutes ahead
python main_es.py predict --n-forecasts 120

# Predict 24 hours ahead (hourly resolution)
python main_es.py predict --resolution 1h --n-forecasts 24
```

E. 30-Day Capacity Planning (Primary Production Command)

Trains on full 1-minute data and projects **30 days into the future**, generating hourly and daily summary tables and charts.

```
# Standard 30-day capacity forecast
python main_es.py capacity

# 30-day capacity forecast using 14 days of history (higher accuracy)
python main_es.py capacity --days 14

# Forecast 60 days ahead instead of 30
python main_es.py capacity --forecast-days 60
```

3. CSV / MySQL Host Pipeline (main.py)

Used for host-level daily metric forecasting from CSV/DB sources.

A. Data Inspection

```
# Show summary statistics for all configured hosts and metrics
python main.py inspect
```

B. Lag Comparison Experiment (Hyperparameter Testing)

Compares model accuracy for different autoregressive lag lookbacks (`n_lags=7` vs `n_lags=14`) to find the optimal lookback.

```
# Compare 7-day vs 14-day lookback performance for a host
python main.py compare-lags --host HYDUPINTAPP16
python main.py compare-lags --host JPRUPIWEBCRP02
```

C. Model Evaluation

Evaluates holdout performance (MAE, RMSE, MAPE) on trailing 30 days of data.

```
# Evaluate host model with 30-day holdout
python main.py evaluate --host HYDUPINTAPP16 --holdout 30
python main.py evaluate --host JPRUPIWEBCRP02 --holdout 30
```

D. Model Training

```
# Fit model for a specific host with chosen lag window
python main.py fit --host HYDUPINTAPP16 --n-lags 7

# Train models for all hosts in parallel
python main.py fit --all --n-lags 7
```

E. Prediction & Forecasting

```
# Forecast 90 days ahead for a host and generate chart
python main.py predict --host HYDUPINTAPP16 --n-lags 7
```

Summary of Output File Locations

Output Type	Elasticsearch Pipeline Location	CSV/MySQL Pipeline Location
Raw Exported Data	outputs/data/es_metrics_<entity_id>.csv / .xlsx	data/processed/*.csv
Capacity Forecast Tables	outputs/forecasts/es/capacity/capacity_hourly_*.csv / .xlsx outputs/forecasts/es/capacity/capacity_daily_*.csv / .xlsx	N/A
Forecast Charts	outputs/forecasts/es/capacity/capacity_hourly_*.png outputs/forecasts/es/capacity/capacity_daily_*.png	outputs/forecasts/forecast_<host>_<met
Saved Models	models/saved/es/model_es_cpu_<entity_id>_<resolution>.pt	models/saved/model_<host>_<metric>.pt